Distance Metrics and Similarity Calculations

https://gemini.google.com/app/dba157b0185c4b3a

User prompt: solve

Response: 1. Distance Measurements

(i) Euclidean Distance

Euclidean distance measures the straight-line distance between two points in an n-dimensional space.

For two points $X = (x_1, x_2, \ldots, x_n)$ and $Y = (y_1, y_2, \ldots, y_n)$, the formula is:

$$d(X, Y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

- It is symmetric: $d(X, Y) = d(Y, X)$.
- It is sensitive to differences in scale across features, so normalization is often recommended before calculation.

(ii) Jaccard Distance

Jaccard distance measures the dissimilarity between two sets (or binary vectors). It is defined as the complement of the Jaccard similarity coefficient:

$$d_J(A, B) = 1 - J(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cup B| - |A \cap B|}{|A \cup B|}$$

For binary attributes where M_{11} is the number of attributes where both objects are 1, M_{01} is where $A = 0, B = 1$, and M_{10} is where $A = 1, B = 0$:

$$d_J = 1 - \frac{M_{11}}{M_{01} + M_{10} + M_{11}} = \frac{M_{01} + M_{10}}{M_{01} + M_{10} + M_{11}}$$

2. Cosine Similarity for Clara and Robert

Given the rating vectors:

- **Clara (C):** $[4.75, 4.5, 5, 4.25, 4]$
- **Robert (R):** $[4, 3, 5, 2, 1]$

The formula for Cosine Similarity is:

$$\text{Cosine Similarity}(C, R) = \frac{C \cdot R}{\|C\| \times \|R\|} = \frac{\sum_{i=1}^{n} C_i R_i}{\sum_{i=1}^{n} C_i^2 \times \quad \sum_{i=1}^{n} R_i^2}$$

Step 1: Compute the dot product ($C \cdot R$):

$$C \cdot R = (4.75 \times 4) + (4.5 \times 3) + (5 \times 5) + (4.25 \times 2) + (4 \times 1)$$

$$C \cdot R = 19 + 13.5 + 25 + 8.5 + 4 = 70$$

Step 2: Compute the magnitude of C ($\|C\|$):

$$\|C\|^2 = (4.75)^2 + (4.5)^2 + 5^2 + (4.25)^2 + 4^2$$

$$\|C\|^2 = 22.5625 + 20.25 + 25 + 18.0625 + 16 = 101.875$$

$$\|C\| = \quad 101.875 \approx 10.0933$$

Step 3: Compute the magnitude of R ($\|R\|$):

$$\|R\|^2 = 4^2 + 3^2 + 5^2 + 2^2 + 1^2$$

$$\|R\|^2 = 16 + 9 + 25 + 4 + 1 = 55$$

$$\|R\| = \quad 55 \approx 7.4162$$

Step 4: Compute the similarity:

$$\text{Cosine Similarity} = \frac{70}{10.0933 \times 7.4162} = \frac{70}{74.854} \approx \mathbf{0.935}$$

3. Euclidean Distance between Angelica and Bill

In collaborative filtering and recommender systems, distance between two users is calculated over the **mutually rated items** (items co-rated by both users).

Mutually rated artists:

1. **Blues Traveler:** Angelica = 3.5, Bill = 2
2. **Broken Bells:** Angelica = 2, Bill = 3.5
3. **Phoenix:** Angelica = 5, Bill = 2
4. **Slightly Stoopid:** Angelica = 1.5, Bill = 3.5
5. **Vampire Weekend:** Angelica = 2, Bill = 3

(Deadmau5, Norah Jones, and The Strokes are omitted because they were only rated by one person.)

Step 1: Compute squared differences:

- Blues Traveler: $(3.5 - 2)^2 = (1.5)^2 = 2.25$
- Broken Bells: $(2 - 3.5)^2 = (-1.5)^2 = 2.25$
- Phoenix: $(5 - 2)^2 = (3)^2 = 9$
- Slightly Stoopid: $(1.5 - 3.5)^2 = (-2)^2 = 4$
- Vampire Weekend: $(2 - 3)^2 = (-1)^2 = 1$

Step 2: Sum the squared differences:

$$\sum (A_i - B_i)^2 = 2.25 + 2.25 + 9 + 4 + 1 = 18.5$$

Step 3: Take the square root:

$$d(\text{Angelica}, \text{Bill}) = \quad 18.5 \approx \mathbf{4.301}$$

4. Relationship Between Minkowski, Euclidean, and Manhattan Distances

Minkowski distance is a generalized metric form in normed vector spaces. For two n-dimensional points X and Y, it is defined as:

$$D(X, Y) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}$$

where p is a parameter ($p \geq 1$).

- **Manhattan Distance (L_1 Norm):** When $p = 1$, the formula simplifies to the sum of absolute coordinate differences (grid/city block distance):

$$D(X, Y) = \sum_{i=1}^{n} |x_i - y_i|$$

- **Euclidean Distance (L_2 Norm):** When $p = 2$, the formula simplifies to the straight-line geometric distance:

$$D(X, Y) = \quad \sum_{i=1}^{n} (x_i - y_i)^2$$

Minkowski distance serves as the overarching generalized framework, with Manhattan distance being its 1st-order case and Euclidean distance being its 2nd-order case.
